

Scalability Analysis of Cross-modal Text-to-Image Pipelines

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Abstract

Cross-modal text-to-image retrieval systems are increasingly deployed over large-scale multimedia corpora, where retrieval effectiveness must be balanced against latency, energy consumption, storage, and candidate fidelity under approximate indexing. In this paper, we present a systems-level evaluation of text-to-image retrieval pipelines under realistic deployment constraints. We study representative single-index, retrieve-and-rerank, and ensemble pipelines on MS-COCO and Flickr30K, and evaluate their scalability on a 7M-image subset of LAION using Flat, HNSW, IVFPQ, and O-IVFPQ indexes. Our results show that strong rerankers can preserve retrieval quality provided the first-stage ANN index maintains sufficient candidate fidelity, but that low-fidelity indexes create unrecoverable candidate losses. At LAION scale, O-IVFPQ provides the strongest single-index operating point, maintaining high candidate fidelity while reducing latency, energy, and storage relative to HNSW. For retrieve-and-rerank pipelines, the reranker dominates end-to-end latency and energy, making index choice less important once sufficient candidate fidelity is preserved. Overall, our study shows that scalable cross-modal retrieval requires joint reasoning about pipeline architecture, candidate fidelity, and systems cost.¹

CCS Concepts

• **Information systems** → **Retrieval efficiency**; **Retrieval effectiveness**.

Keywords

cross-modal retrieval; text-to-image retrieval; retrieval efficiency; index selection

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¹Code: <https://github.com/data-iitd/mmir-cikm>



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1 Introduction

As information increasingly spans multiple modalities—text, images, tables, audio, and video—the need for systems that enable cross-modal retrieval has become paramount. Among these, text-to-image retrieval is one of the most widely studied and practically deployed tasks: it allows users to describe an image in natural language and retrieve matching visual content. Applications include multimedia search engines, product recommendation, medical diagnostics, and generative AI pipelines where images are grounded in language.

Recent prior work in multimodal retrieval, in the age of transformers, has aimed to train both efficient and accurate models. Typically, cross-encoders such as BLIP-2 [16] and Qwen3-VL-Reranker [18] are expected to be more accurate, while bi-encoder models such as CLIP [24], FLAVA [28], and OpenCLIP [2], that unify text and image representations, are faster since the corpus can be pre-processed and indexed. Late interaction models such as LightningDOT [30], llama-nemoretriever-colembed [32], and ColPali [7] aim to balance both. Other models, such as instruction-tuned models [31], are also candidates for cross-modal retrieval tasks.

Apart from individual models, other pipelines (long studied in IR) can also be envisioned. For example, multi-stage “retrieve and rerank” cascades where a (fast) first stage generates candidates, that subsequent stages rerank (see [19] for a discussion on the history of such pipelines), fusion strategies to rank results from two separate indexes [3, 25], etc., can also be envisioned.

Despite rapid progress in these retrieval models, most existing evaluations focus primarily on effectiveness (accuracy metrics such as $R@k$) on relatively small datasets such as MS-COCO [20] and Flickr30K [33]. While these benchmarks are important for comparing retrieval quality, they are not useful for studying deployment constraints for large-scale retrieval. Specifically, query latency, energy consumption, and storage are important metrics for retrieval systems with millions of images. Further, at such scales, exact KNN algorithms are infeasible, and so, we need to consider Approximate Nearest Neighbor (ANN) algorithms and indexing.

Given this, any retrieval pipeline, such as a retrieve-and-rerank pipeline will depend critically on the *candidate fidelity* of the candidate set generated by the first-stage retriever. ANN indexing not only impacts candidate fidelity and consequently, downstream retrieval quality, but also latency and memory usage. Therefore, *retrieval pipelines must be studied along with the indexing strategy used to deploy them.*

In this paper we study exactly this question: **How do cross-modal retrieval pipelines and ANN indexing strategies interact under realistic deployment constraints?**

1.1 Scope of our Study

To address the above question, we perform a systematic systems-level evaluation of text-to-image pipelines spanning: i) efficient single-index retrieval, ii) multi-stage retrieve-and-rerank architectures, and iii) ensemble retrieval pipelines.

We evaluate these pipelines using both exact and approximate indexing strategies, including, HNSW [22], IVFPQ [1, 10], and O-IVFPQ [9] across multiple metrics, including retrieval effectiveness, latency, energy consumption, storage footprint, and candidate fidelity under ANN approximation.

Our evaluation combines benchmark-scale retrieval experiments on MS-COCO [20] and Flickr30K [33] with large-scale indexing experiments on a 7M-image subset of LAION-400M [27]. Rather than exhaustively benchmarking every model configuration, our goal is to identify representative operating points spanning different regions of the effectiveness-efficiency trade-off.

Our experiments yield several important observations. First, the retrieval architecture has a substantial impact on retrieval quality, rather than simply the individual models used. Second, candidate recall saturates rapidly for many retrievers, suggesting that approximate indexing can preserve sufficient candidate fidelity for downstream re-ranking. Third, among the indexing strategies studied, O-IVFPQ provides the strongest overall effectiveness-efficiency tradeoff – that is, it maintains good candidate fidelity while reducing storage requirements and latency, compared to the alternatives.

1.2 Contributions

In summary, our contributions are the following.

- We present a systems-level evaluation of cross-modal retrieval pipelines under realistic deployment constraints, spanning retrieval effectiveness, latency, energy consumption, and storage.
- We analyse how ANN indexing strategies affect candidate fidelity and identify representative deployment operating points across single-index, retrieve-and-rerank, and ensemble pipelines.
- We provide practical insights into the trade-offs of deploying these cross-modal pipelines.

Organization. The rest of this paper is organized as follows. In Section 2 we describe the class of models and pipelines that we study. The evaluation setup for our experiments is described in Section 3 and Section 4. We present our evaluation in Section 5. We summarize related work in Section 6 and conclude in Section 7.

2 Retrieval Pipelines

Modern text-to-image retrieval systems can be instantiated in several different ways depending on the desired tradeoff between retrieval quality, latency, memory usage, and scalability. In this paper, we select a representative set of pipelines that can each be instantiated with different models. Our goal is not to exhaustively measure the effectiveness of specific models, but to instead study

how retrieval pipelines can be composed and their scalability on large-scale data.

2.1 Models

A variety of models are available for instantiating retrieval pipelines. We classify them as follows:

Bi-encoders: A bi-encoder independently encodes queries and documents into a shared embedding space, thus allowing documents to be indexed. For a given query, top- k retrieval involves finding the k nearest neighbors in the document index. In the text-only setting DPR [14] exemplifies the use of bi-encoders in retrieval. In the text-to-image setting, several models such as CLIP/OpenCLIP [24], FLAVA [28], UniIR [31], etc. are all bi-encoders.

Cross-encoders: Cross-encoders jointly process $\langle \text{query}, \text{document} \rangle$, allowing query and document tokens to interact with each other. BERT [5] is the canonical example from the text world. In the cross-modal setting, we have models such as BLIP-2 [16] and Qwen3-VL [18].

Late-interaction models: Late-interaction models are similar to bi-encoders in that they allow documents to be processed offline and indexed. However, they store multiple tokens per document. Token-level interaction between query tokens and document tokens take place at query time. Thus, these models occupy an intermediate point – more accurate than bi-encoders and more efficient than cross-encoders. ColBERT [15] is the canonical example from the text retrieval task. In our setting, examples include llama-nemoretriever-colembed (llama-nc) [32] and ColPali [7].

Instruction tuned models: Instruction-tuned models augment the query with an instruction ("*Find an image of a dog playing with a ball*"), which is useful when many modalities are involved and retraining is expensive. In our setting, we experimented with [31].

2.2 Pipeline families

Pipeline family	Representatives	Expected behaviour
Single-index	CLIP, OpenCLIP, llama-nc	Low-latency retrieval
Retrieve-and-rerank	OpenCLIP $\rightarrow$ Qwen3-VL, UniIR $\rightarrow$ BLIP-2	High-accuracy retrieval
Ensembles	OpenCLIP + UniIR	Complementary retrieval

Table 1: Summary of pipeline families and examples of how they could be instantiated.

Single index retriever: Single-index pipelines directly retrieve the top- k images for the given text query. The index is created with either a bi-encoder, late-interaction or instruction-tuned model.

Retrieve-and-rerank: This pipeline involves two stages. In the first stage, a fast single-index retriever generates candidates. In the second stage, a more accurate, but slower reranker (typically, a cross-encoder) reranks the candidates.

Ensembles: Ensemble pipelines combine candidates from multiple retrievers using Reciprocal Rank Fusion (RRF) [3]. These

pipelines attempt to recover complementary retrieval failures across models.

3 Metrics and Methodology

Cross-modal retrieval systems must typically operate over corpora consisting of millions or billions of data items. At this scale, retrieval accuracy is only one of the metrics useful for evaluating pipeline quality. Metrics such as latency, storage footprint and energy consumption become equally important.

To study scalable deployment settings, we therefore evaluate retrieval systems in two ways: i) benchmark-driven retrieval accuracy on MS-COCO [20] and Flickr30K [33], and, ii) scalability metrics on a 7 million image subset of LAION-400M [27]. Rather than exhaustively benchmarking pipeline instantiations with a multitude of models, we instead focus on representative operating points. We describe our metrics and methodology in more detail below.

3.1 Datasets

In order to measure retrieval accuracy, we use the test splits of MS-COCO (5,000 queries) and Flickr30K (1,000 queries) from [13]. Each image is associated with 5 captions. One of these captions is used as a query, and that query has a single ground-truth image. Due to space constraints, we focus on MS-COCO, but the trends reported for MS-COCO were also observed for Flickr30K.

To study large-scale retrieval, we use a 7M subset of LAION-400M [27]. Unlike MS-COCO and Flickr30K, LAION-scale retrieval allows us to study how indexing strategies affect latency, energy consumption, and storage footprint.

3.2 Effectiveness and Candidate Fidelity

On Flickr30K and MS-COCO, retrieval effectiveness is measured using Recall@ k (R@ k). The retrieval is done through an *exact nearest neighbor* algorithm. So, for each pipeline instantiation that we present later, the recall numbers represent a *ceiling* on the accuracy.

For large-scale evaluation with LAION-7M, retrieval is feasible only with ANN algorithms. Further, due to a lack of robust ground-truth information in this dataset, we measure effectiveness using *candidate fidelity* (referred to as *CF* or just *fidelity*). That is, the average overlap between the exact top- k and approximate top- k over a set of random queries Q .

$$CF@100 = \frac{1}{|Q|} \sum_{q \in Q} \frac{|\text{Top}100_{\text{KNN}}(q) \cap \text{Top}100_{\text{ANN}}(q)|}{100},$$

Candidate fidelity is not a relevance metric, rather, it measures how faithfully an approximate index preserves the candidate set available to downstream reranking or fusion stages. Candidate fidelity changes based on the index used and its parameters (see Section 4.) Apart from the accuracy metrics, all other metrics described below use only LAION-7M.

3.3 Latency

We decompose per-query latency as

$$T_{\text{total}} = T_{\text{embedding}} + T_{\text{retrieval}} + T_{\text{rerank}} + T_{\text{overhead}} \quad (1)$$

where $T_{\text{embedding}}$ is the encoder forward pass on the query, $T_{\text{retrieval}}$ is the FAISS [12] index lookup, T_{rerank} is the optional cross-encoder pass over the candidate set, and T_{overhead} accounts for inter-stage

data movement. We report the median per-query T_{total} over the full query set, with each stage broken out separately so that the dominant cost regime is visible at a glance.

3.4 Energy

Per-query GPU energy is measured per batch in joules via the NVIDIA Management Library (NVML) and normalised to a per-query value. We report energy at the same per-stage granularity as latency so the two metrics can be cross-referenced: in single-index pipelines the encoder and index together account for the bulk of energy, whereas in retrieve-and-rerank pipelines the cross-encoder dominates by two to three orders of magnitude.

3.5 Storage

We report on-disk storage utilisation on LAION-7M, including both the serialised FAISS index and the associated metadata required at query time. Storage figures do not vary across encoders—every model in our study produces 768-dimensional embeddings stored as 4-byte floats, so the raw embedding payload is identical—but they do vary across indexing algorithms, because the underlying structures differ. HNSW stores the navigation graph in addition to the uncompressed vectors, IVFPQ stores only PQ codes and cluster centroids, and Flat stores the raw vectors. Accuracy-optimised configurations therefore typically carry a different footprint than their default counterparts—most notably, O-IVFPQ with PQ256 codes is larger than standard IVFPQ with $m=96$, but remains an order of magnitude smaller than HNSW.

3.6 Hardware setup

All experiments were run on a single A100 GPU (40GB VRAM) and 128GB system RAM.

4 Scalability: Operating Points

Evaluating retrieval scalability at the LAION-7M scale is non-trivial because the choice of indexing strategy and its parameters jointly determine accuracy, latency, energy, and storage. Recall that exact search (which scales linearly with corpus size) is not feasible for large datasets. Therefore, all scalability experiments are performed with index structures that utilize approximate nearest neighbor (ANN) algorithms. The options explored are described next.

Approximate search options. Two ANN families dominate practical deployments and represent qualitatively different design choices:

HNSW [22]: a hierarchical navigable small-world graph supporting sub-linear query-time traversal. HNSW typically achieves very low query latency and high accuracy, but the graph is held entirely in memory and incurs a large storage footprint. FAISS does not support GPU execution for HNSW, so all HNSW queries run on CPU.

IVFPQ [1, 10]: partitions the vector space via k -means into n list Voronoi cells and applies product quantisation (PQ) [10] to compress stored vectors into compact codes. At query time only n probe cells are searched. IVFPQ yields the smallest on-disk footprint and the lowest per-query energy, at the cost of additional approximation from quantisation. An **OPQ**-rotated variant (O-IVFPQ) [9] applies an optimised orthogonal rotation before quantisation

to better preserve nearest-neighbour structure. Flat and IVFPQ execute on GPU via FAISS; for the exact reference in the fidelity-optimised comparison (Table 8, Figure 6) we additionally measure Flat executed on CPU.

4.1 Operating Points

Because exhaustive search over the joint parameter space is impractical at 7 M scale, we ran a structured parameter sweep in which we swept one parameter at a time around a central operating point. Table 2 summarises the grids we explored and the best fidelity achieved within each sub-grid, while Figure 1 visualises the parameter sweeps and their effect on candidate fidelity (see Section 3.2). We next discuss how these various indexes behave under these different sets of parameters.

4.1.1 Behaviour of ANN Indexes. Figure 1 shows how candidate fidelity changes as the primary search parameters are varied for each ANN family. Each curve corresponds to one build configuration – the x -axis sweeps a search parameter. We observe the following behaviour for each type of index.

HNSW improves steadily with increasing $efSearch$ because larger search budgets allow the traversal of more of the navigation graph and therefore recover more of the exact nearest neighbor structure. Candidate fidelity is highest (91%) for $M = 48$, after which performance saturates and even slightly declines for larger M values. This may be because there is no consistent improvement in neighborhood quality despite a denser graph.

In contrast, IVFPQ exhibits flat behaviour with increasing $nprobe$. This is likely because the dominant source of retrieval degradation is because of the distortion introduced by product quantisation (which OPQ tries to correct). Consequently, the index can only recover its neighborhood information in a very limited way by increasing $nprobe$.

O-IVFPQ, which uses OPQ [9] rather than standard PQ to compress vectors behaves differently and indeed improves candidate fidelity substantially. The rotation-before-quantisation method of OPQ redistributes variance across subspaces through vector rotation. Therefore, it reduces distortion and retains more neighborhood information.

Default vs. fidelity-optimised configurations. For our main efficiency measurements (Section 5.2) we adopt a single *default* configuration per family corresponding to representative deployment choices in the FAISS ecosystem [12]: $M=32$, $efSearch=128$ for HNSW [22], and $nlist=4096$, $m=48$, $nprobe=8$ for IVFPQ [1, 10]. These values are not optimised for our data; they serve as a baseline against which the accuracy-optimised configurations discovered by the ablation (Section 5.3.1) are compared.

5 Evaluation

We evaluate the retrieval pipelines described in Section 2.2 along three research questions:

- **RQ1:** How does the pipeline design affect retrieval accuracy?
- **RQ2:** What are the efficiency characteristics of these pipelines at scale?
- **RQ3:** What are the trade-offs between efficiency and accuracy? This is the key question we answer in this paper.

HNSW			
M	efConstruction	efSearch	Best Fidelity (%)
16	{100, 200}	{128, 256, 512, 1024, 2048}	79.82
32	{100, 200}	{128, 256, 512, 1024, 2048}	84.92
48	{400}	{128, 256, 512, 1024, 2048}	90.73
64	{200, 400}	{128, 256, 512, 1024, 2048}	85.04
64	{800}	{1024, 2048, 4096}	87.43
80	{1200}	{1024, 2048, 4096}	74.05

IVFPQ			
m	nprobe	nlist	Best Fidelity (%)
16	{32, 64, 128, 256}	{2048, 4096, 8192}	15.43
32	{32, 64, 128, 256}	{2048, 4096, 8192}	24.16
64	{32, 64, 128, 256}	{4096, 8192}	38.78
96	{512, 1024, 2048, 4096}	{16384, 32768}	53.22

O-IVFPQ			
OPQ/PQ	nlist	nprobe	Best Fidelity (%)
192	16384	{1024, 2048, 4096, 8192, 16384, 32768}	79.62
192	32768	{1024, 2048, 4096, 8192, 16384, 32768}	80.01
256	16384	{1024, 2048, 4096, 8192, 16384, 32768}	83.51
256	32768	{1024, 2048, 4096, 8192, 16384, 32768}	83.77

Table 2: Parameter sweeps over each ANN family on LAION-7M. Each row fixes one identifying parameter (M for HNSW, m for IVFPQ, the OPQ/PQ code-word dimension for O-IVFPQ) and reports the best candidate fidelity (CF@100) achieved by sweeping over the remaining parameters in the listed ranges. Bold fidelity values mark the configurations carried forward as the fidelity-optimised operating points in Section 5.3.1.

RQ1 is addressed in Section 5.1 through experiments on MS-COCO and Flickr30K. RQ2 is addressed in Section 5.2 through large-scale experiments on a 7M-image subset of LAION-400M. RQ3, the central question of this paper, is addressed in Section 5.3 by jointly considering the findings from the first two.

5.1 Pipeline Accuracy

Table 3 shows the accuracy of several representative instantiations of the pipelines discussed in Section 2.2 with different models.

The main pattern to note here is that re-ranking gives big gains in the recall numbers. For example, the $R@1$ of UniIR_{joint} increases by about 10% with Qwen3-VL reranker and 17% with BLIP-2 (expectedly so, since cross-encoders are considerably more powerful). Further, ensembles remain competitive with cross-encoder rerankers at higher recall levels, but do not match the cross-encoder at $R@1$.

5.1.1 Candidate Recall Saturation. Figure 2 shows how Recall@ k changes as k increases for representative single-index retrievers (which are also first-stage retrievers in the retrieve-and-rerank pipeline). As can be seen, the retrievers all achieve over 95% at $k = 100$ and over 90% at $k = 50$. This implies that the relevant item has been most likely retrieved within relatively small candidate sets, even before any reranking.

Once candidate recall saturates, downstream rerankers operate over highly overlapping candidate pools even when the first-stage retrievers differ substantially. Consequently, the reranker increasingly determines the final ranking quality and the first-stage retriever becomes less important.

This observation implies that an efficient first-stage retriever can be selected provided candidate recall remains high. More importantly, this observation also motivates the use of approximate

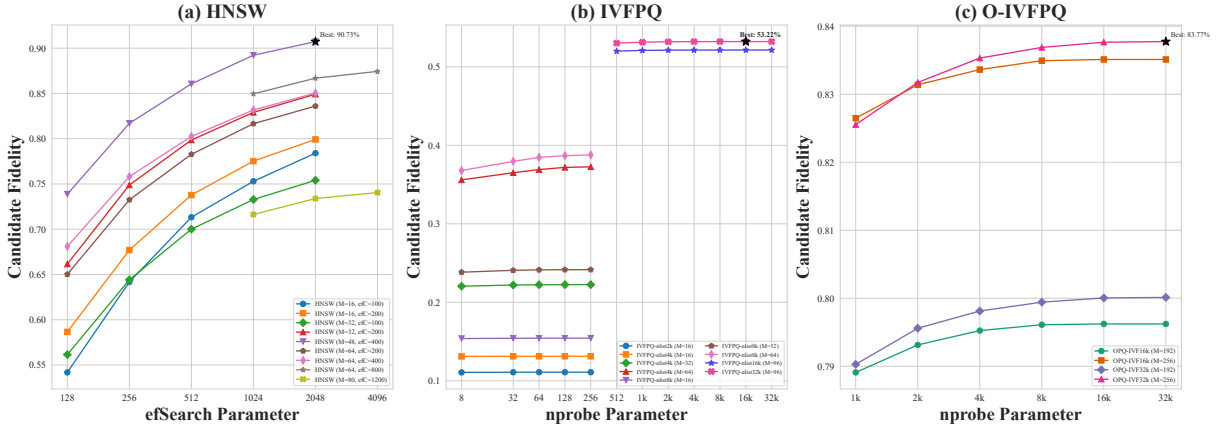


Figure 1: Candidate Fidelity vs. the swept search parameter for (a) HNSW, (b) IVFPQ, and (c) O-IVFPQ on LAION-7M, one line per build configuration. HNSW improves with efSearch (best at $M=48$); IVFPQ is flat in nprobe and separated by code length m ; O-IVFPQ’s rotation lifts candidate fidelity to 78–84%. Best per-family fidelity is marked.

Single index	llama-nc			UniR _{Joint}			ColPali		
	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10
	0.5546	0.7716	0.847	0.5534	0.7592	0.8384	0.4852	0.7242	0.8042
Retrieve and re-rank with Qwen3-VL cross-encoder reranker (100 candidates in the Retrieve stage)	UniR _{Joint} → Qwen3-VL			OpenCLIP _{Image} → Qwen3-VL			UniR _{Image} → Qwen3-VL		
	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10
	0.6104	0.7948	0.8622	0.5662	0.803	0.8754	0.5652	0.8052	0.8776
Retrieve and re-rank with BLIP-2 cross-encoder reranker (100 candidates in the Retrieve stage)	UniR _{Joint} → BLIP-2			OpenCLIP _{Image} → BLIP-2			CLIP _{Image} → BLIP-2		
	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10
	0.6498	0.7621	0.8230	0.6496	0.8564	0.9110	0.6494	0.8525	0.9059
Ensemble (top-50 candidates from each model)	(FLAVA _{Image} , UniR _{Joint})			(CLIP _{Image} , UniR _{Joint})			(OpenCLIP _{Image} , UniR _{Image})		
	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10
	0.5446	0.8102	0.8760	0.5368	0.7946	0.8530	0.5262	0.7844	0.8598

Table 3: Recall numbers for the MS-COCO dataset. The subscript $_{Image}$ indicates image embedding for that model, $_{Joint}$ indicates a joint $\langle query, image \rangle$ embedding specific to UniR.

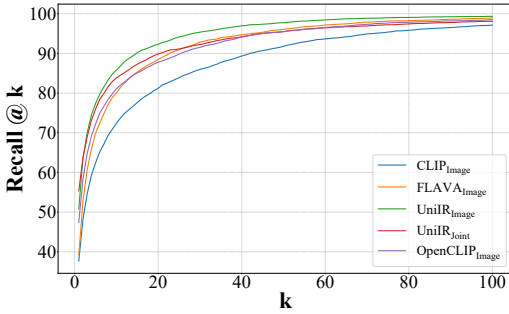


Figure 2: Recall@k vs. k for various models.

nearest neighbour indexing that we study next – ANN-based retrieval is viable, provided sufficient candidate fidelity is preserved for downstream reranking or fusion-based ensemble pipelines. Note that although Recall@k (candidate recall for a first-stage retriever) and candidate fidelity are different metrics, both quantify whether relevant items survive into downstream reranking stages.

5.1.2 Accuracy Degradation with ANN. Table 4 shows how R@1 changes for representative retrieve-and-rerank pipelines when the

first-stage retriever is replaced with various ANN indexes under different operating points. While ANN indexing reduces candidate fidelity relative to exact nearest neighbour retrieval, the downstream reranker is able to recover most of the lost retrieval effectiveness provided that the candidate set remains sufficiently faithful to the exact top-k retrieval.

In particular, we observe that moderate reductions in candidate fidelity (70%–80% CF) result in only small reductions in end-to-end R@1. This suggests that retrieve-and-rerank pipelines do not require exact nearest-neighbour retrieval at the first stage; instead, they require only that the relevant item remains present in the retrieved candidate set. The same argument can be made for ensemble pipelines as well, since they too depend on the presence of the relevant candidates in the retrieved set.

However, once candidate fidelity drops substantially (to less than 70% CF), retrieval effectiveness degrades more sharply. Since the reranker can only reorder candidates retrieved by the ANN stage, relevant items omitted during approximate retrieval cannot be recovered downstream. Therefore, candidate fidelity acts as an upper bound on the retrieval effectiveness achievable by downstream re-ranking. Overall, these results support the use of ANN indexing for scalable cross-modal retrieval systems. Exact nearest-neighbour retrieval is not necessary for maintaining competitive end-to-end

Encoder	Tier	Index	Configuration	Fidelity (CF@100)	R@1	R@1 ↓
CLIP _{Image}	–	Flat (exact KNN)	Full precision	100%	0.6494	–
	Moderate Fidelity	HNSW	$M=32$, efC=200, efSearch=128	96.46%	0.6406	-1.36%
		IVFPQ	$m=48$, nprobe=16, nlist=128	71.14%	0.6160	-5.14%
		O-IVFPQ	OPQ64, IVF128, PQ48, nprobe=16	73.90%	0.6212	-4.34%
	Low Fidelity	HNSW	$M=8$, efC=40, efSearch=16	50.20%	0.5064	-22.02%
		IVFPQ	$m=48$, nprobe=1, nlist=8	59.98%	0.4816	-25.84%
		O-IVFPQ	OPQ64, IVF8, PQ48, nprobe=1	61.90%	0.4798	-26.12%
OpenCLIP _{Image}	–	Flat (exact KNN)	Full precision	100%	0.6496	–
	Moderate Fidelity	HNSW	$M=16$, efC=100, efSearch=64	92.19%	0.6434	-0.95%
		IVFPQ	$m=48$, nprobe=16, nlist=128	75.65%	0.6344	-2.34%
		O-IVFPQ	OPQ64, IVF128, PQ48, nprobe=16	78.56%	0.6350	-2.25%
	Low Fidelity	HNSW	$M=8$, efC=40, efSearch=16	55.76%	0.5762	-11.30%
		IVFPQ	$m=48$, nprobe=1, nlist=8	66.23%	0.5210	-19.80%
		O-IVFPQ	OPQ64, IVF8, PQ48, nprobe=1	67.85%	0.5214	-19.74%

Table 4: KNN vs. ANN comparison for the retrieve-and-rerank pipeline (encoder → BLIP-2) on MS-COCO using index configurations that result in moderate and low fidelity. R@1 is end-to-end recall after reranking; R@1 ↓ is the relative reduction in R@1 from the exact-KNN baseline.

retrieval quality, provided that the ANN index preserves sufficiently high candidate fidelity.

5.1.3 Implications for Scalable Retrieval. From the previous analysis, we derive the following guidelines for scalable retrieval.

(1) Strong rerankers reduce sensitivity to the choice of first-stage retriever. (2) Candidate recall saturates rapidly, making ANN-based retrieval viable for large-scale deployment. (3) ANN quality matters mainly to ensure that sufficient candidate fidelity is maintained for downstream ranking.

These observations now motivate the large-scale scalability analysis, including, latency, energy consumption and storage for indexing strategies in Section 5.2 (RQ2) and then the accuracy-vs-scalability tradeoff (RQ3) in Section 5.3.

5.2 Scalability Analysis (RQ2)

We now turn to efficiency at scale. All experiments in this section use the 7-million-image LAION subset described in Section 3.1. The indexing strategies, pipeline configurations, and efficiency metrics are described in Section 4. We summarise the key results below.

5.2.1 Latency Analysis. Figure 3 reports median per-query latency (seconds, log scale) broken down by index type and pipeline stage². Encoding time is shown as a hatched component. The left panel covers single-index and retrieve-and-rerank pipelines; the right panel (after the dashed line) covers ensembles. In addition, Tables 5 and 6 report the query-level latency variability (the median, 25th and 75th percentiles)³.

²The median latency was computed as follows: 3 runs of the queries were performed, and the median for each run was computed. The reported number is the median of these 3 medians.

³These are not confidence intervals, but summarize how latency varies.

Single-index pipelines. IVFPQ is the fastest index across all models, achieving 13–17 ms per query depending on the encoder. HNSW is moderately slower at 24–28 ms, while Flat Index requires 51–55 ms. Encoding time (8–12 ms) is consistent across index types for a given model, confirming that the encoder forward pass is independent of the downstream index. Notably, for IVFPQ, the index lookup itself takes <5 ms; encoding dominates end-to-end latency. *Further latency reductions would require model-side optimisations* (quantisation, distillation, or embedding caching) rather than index-side improvements.

Retrieve-and-rerank pipelines. Adding a cross-encoder reranking stage dramatically increases latency. With BLIP-2, all configurations exhibit ~4.5 s per query – a *two-to-three orders-of-magnitude increase* over single-index retrieval – dominated entirely by the cross-encoder’s inference over 100 candidates. All three index structures fall within a narrow band (± 0.15 s), confirming that **the reranker is the dominant latency bottleneck; the choice of index structure is essentially irrelevant for efficiency once a cross-encoder reranking is applied.**

Qwen3-VL presents a different picture at ~1.1 s – approximately 4× **faster** than BLIP-2. This speedup is consistent across all retrievers and indexes, suggesting it is an intrinsic property of the Qwen3-VL architecture – expectedly, because of the size of the model. The ~1.1 s response time is feasible for many user-facing applications where 4.5 s would not be.

Ensemble pipelines. Latency of ensemble pipelines is along expected lines: with two retrievers executed in sequence, per-query latency is roughly the sum of the two constituent single-index pipelines plus a small fusion overhead (≈ 15 ms). HNSW-based ensembles achieve 65–70 ms per query; IVFPQ ensembles are faster at roughly 42–46 ms across all six pairwise combinations (Figure 3);

Pipeline	Index	Encoding (ms)	Retrieval (ms)	Reranking (s)
CLIP → BLIP-2	HNSW	8.3 [7.6, 8.5]	24.0 [21.7, 25.9]	4.53 [4.39, 4.68]
	IVFPQ	8.5 [7.9, 8.8]	13.2 [12.2, 13.8]	4.54 [4.43, 4.67]
	Flat	8.2 [7.7, 8.6]	50.8 [50.2, 51.2]	4.50 [4.39, 4.62]
CLIP → Qwen3-VL	HNSW	8.3 [7.6, 8.5]	24.0 [21.7, 25.9]	1.05 [1.01, 1.10]
	IVFPQ	8.5 [7.9, 8.8]	13.2 [12.2, 13.8]	1.05 [1.02, 1.10]
	Flat	8.2 [7.7, 8.6]	50.8 [50.2, 51.2]	1.05 [1.02, 1.09]

Table 5: Per-stage query latency for retrieve-and-rerank pipelines on LAION-7M; each cell reports the median with [p25, p75]. Encoding and retrieval are identical across rerankers (same encoder and index); the reranker dominates total latency.

Table 6 reports query-level variability for four representative pairs. We do not report KNN ensembles: the Flat index runs on GPU (Section 4), and at 21.25 GB per index (Figure 5) loading two indexes onto GPU simultaneously would exceed the 40 GB VRAM on our hardware.

5.2.2 Energy Consumption. Figure 4 reports GPU energy consumption per query (J, log scale), measured via NVML.

Single-index pipelines. IVFPQ is the most energy-efficient index at 0.60–0.78 J per query, roughly 1.5× lower than HNSW (0.96–1.2 J) and 8–12× lower than Flat (6.4–7.6 J). The large Flat gap reflects the cost of exhaustive GPU computation over 7M vectors.

Retrieve-and-rerank pipelines. Reranking with BLIP-2 amplifies energy to ~380 J per query – two to three orders of magnitude above single-index retrieval. The reranker dominates so completely that the index choice is irrelevant for energy budgeting. **Optimising the index structure yields negligible energy savings when cross-encoder reranking is applied.** Qwen3-VL consumes approximately **half the energy of BLIP-2** at ~185 J per query, making it the more sustainable choice when reranking is desired.

Ensemble pipelines. HNSW-based ensembles consume 2.2–2.4 J; IVFPQ ensembles consume only 1.4–1.6 J – comparable to a single HNSW query. Ensemble energy is almost exactly the sum of its two constituent single-index pipelines (a single IVFPQ pipeline costs 0.60–0.78 J, so two predict 1.2–1.6 J against 1.4–1.6 J measured), confirming that reciprocal rank fusion itself contributes negligible energy: the entire cost is the two retrieval passes. Because the paired models use distinct encoders, both the encoder forward pass and the index lookup are incurred twice. **IVFPQ ensembles deliver multi-model fusion at roughly 1.5× lower energy than HNSW ensembles, and at a total cost that remains more than two orders of magnitude below cross-encoder reranking.**

5.2.3 Storage Footprint. Figure 5 reports on-disk storage at LAION-7M for twelve representative configurations across the three approximate index families and the exact reference. Three trends are immediate.

HNSW does not meaningfully compress. The graph adds at most ~2.5 GB over Exact’s 21.25 GB raw-vector payload, ranging from 21.22 GB at $M=16$ to 23.72 GB at $M=64$. HNSW’s accuracy advantage in Table 8 therefore comes at essentially the full raw-embedding storage cost.

Ensemble	Index	Total latency (ms)
(CLIP, FLAVA)	HNSW	68.0 [64.2, 72.3]
	IVFPQ	42.9 [40.8, 45.0]
(CLIP, UniIR)	HNSW	66.7 [62.3, 72.6]
	IVFPQ	40.9 [39.0, 43.5]
(OpenCLIP, CLIP)	HNSW	64.3 [60.2, 68.6]
	IVFPQ	42.5 [39.9, 45.2]
(OpenCLIP, UniIR)	HNSW	66.2 [61.1, 70.7]
	IVFPQ	42.7 [40.4, 45.6]

Table 6: Total per-query latency for four ensemble pipelines on LAION-7M (median with [p25, p75]).

The reason for HNSW’s large storage footprint is because of the structure of the HNSW index itself. HNSW stores *both* the original vectors and a multi-layer navigation graph. The connectivity of the navigation graph grows with M and $efConstruction$ parameters. The additional graph links improve traversal quality and therefore, candidate fidelity. But this is at the cost of higher memory consumption.

Standard IVFPQ is the strongest compressor. It ranges from 0.41 GB ($m=16$) to 1.02 GB at the accuracy-optimised configuration ($m=96$, Table 7)—roughly 22× smaller than HNSW at the same operating point. Within the family, storage scales with the sub-quantiser count m , consistent with each vector being encoded as m bytes (one byte per sub-quantizer at $nbits=8$).

O-IVFPQ buys accuracy with a ~2× storage premium over IVFPQ. OPQ-rotated variants occupy 1.60–2.15 GB depending on code-word dimension and $nlist$, roughly twice the footprint of standard IVFPQ but still an order of magnitude below HNSW. At 2.15 GB the fidelity-optimised configuration delivers 83.77% candidate fidelity (Table 8) at only ~10% of HNSW’s on-disk size.

The reason for this large fidelity improvement is because OPQ [9] rotation redistributes variance across subspaces before quantisation. This reduces distortion compared to PQ [10]. Therefore, the nearest neighbor structure is better preserved by OPQ.

Storage therefore reinforces the operating-point ordering established by latency and energy: $IVFPQ < O-IVFPQ \ll HNSW \approx \text{Exact}$.

5.3 Fidelity–Efficiency Trade-offs (RQ3)

Table 8 consolidates latency, energy, candidate fidelity, and storage for every pipeline-index combination we evaluate at LAION-7M scale. *Single-index* rows use each family’s fidelity-optimised configuration (Section 5.3.1); *ensemble* and *reranker* rows use the default configurations of Section 4. Candidate fidelity is reported only for the first stage retriever since rerankers or fusions operate over retrieved candidates rather than producing any approximate set of candidates. Together with the benchmark accuracy results in Section 5.1, this table answers the central question of the paper: given a latency, energy, or storage budget, which indexing strategy and pipeline should a practitioner deploy?

5.3.1 Default and Fidelity-Optimised Index Configurations. We evaluate every index family in two operating regimes. The *default* regime uses the parameter settings introduced in Section 4 and is the basis for the latency, energy, and storage measurements reported in Section 5.2. The *fidelity-optimised* regime uses the best

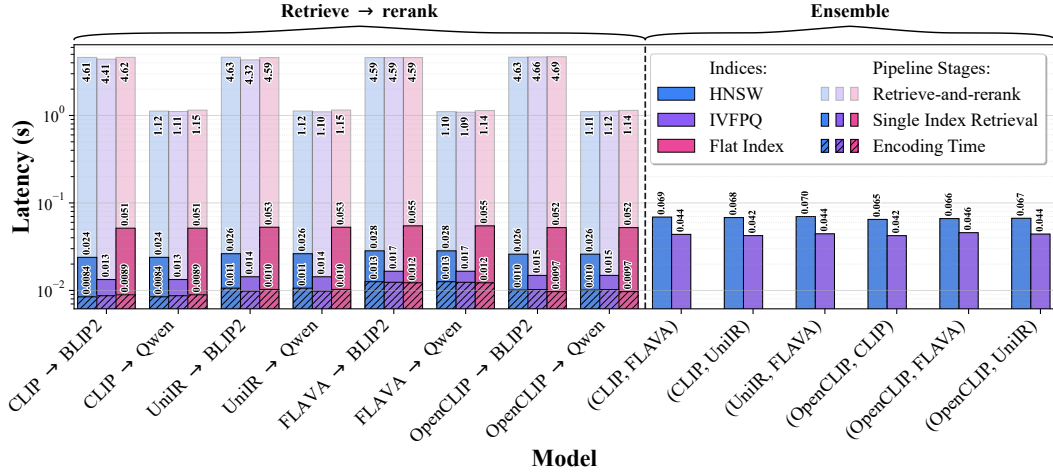


Figure 3: Median per-query latency (log scale) on LAION-7M. Left: single-index and retrieve-and-rerank. Right (after dashed line): ensembles. Hatched regions indicate encoding time.

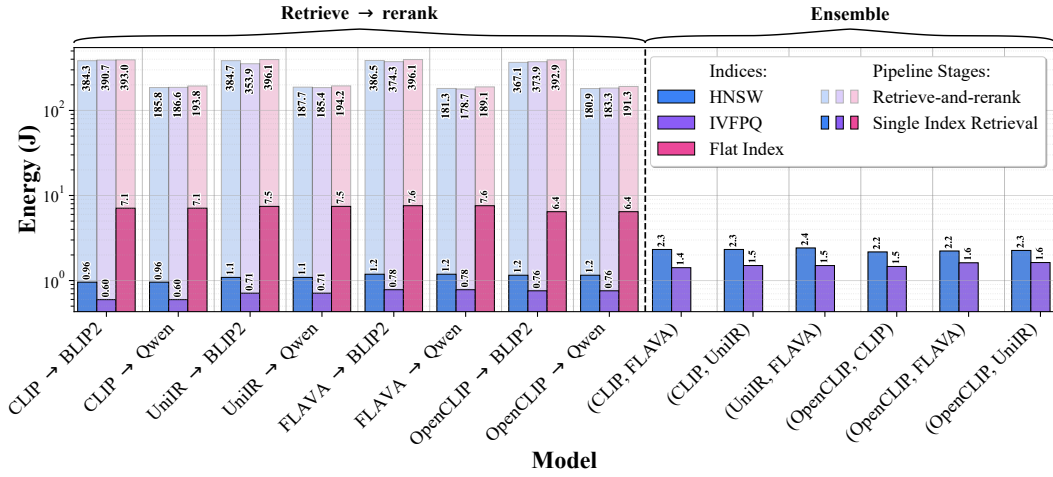


Figure 4: Per-query GPU energy consumption (J, log scale) on LAION-7M, measured via NVML over a batch of 1000 queries and normalised per query. Left: single-index and retrieve-and-rerank. Right (after dashed line): ensembles.

Index family	Fidelity	Best configuration
HNSW	90.73%	$M=48$, efC=400, efSearch=2048
IVFPQ (standard)	53.22%	$m=96$, nprobe=4096, nlist=32768
O-IVFPQ	83.77%	OPQ256, IVF32768, nprobe=32768

Table 7: Fidelity-optimised configurations selected from the parameter sweep grids in Table 2. Fidelity is measured over the full LAION-7M query set.

configuration discovered by our parameter sweeps over the grids in Table 2, and characterises the upper end of each family’s accuracy frontier. Table 7 lists the three fidelity-optimised configurations carried forward.

The fidelity gap between regimes is quite large, particularly for IVFPQ: the standard parameters recover only $\sim 53\%$ of the exact top-100, while the OPQ-rotated variant with longer codes lifts this to nearly 84%. HNSW peaks at 90.73% at $M=48$ and does not improve in the high-build grid ($M \geq 64$), consistent with the hub-formation effects discussed in Section 4. Note that the reason for the relatively low performance of IVFPQ is that the main source of error is because of the quantization method itself. Increasing $nprobe$ in this case provides very limited gains because of this – that is, if the vector reconstruction quality is not good enough, then simply searching more inverted lists may not recover the nearest-neighbors. OPQ attempts to fix this problem – lowers the distortion in the quantization.

5.3.2 Findings. Note the apparent latency gap between Section 5.2.1 and Table 8: the single-index numbers here correspond to the

Pipeline family	Index	Latency	Energy	Fidelity	Storage	Config
Single Index	IVFPQ	151 ms	5.15 J	53.22%	1.02 GB	$m=96$, $nprobe=4096$, $nlist=32768$
	HNSW	96 ms	3.15 J	90.73%	22.89 GB	$M=48$, $efC=400$, $efSearch=2048$
	KNN (CPU)	6.45 s	219.7 J	100%	21.25 GB	Full Precision
	O-IVFPQ	54 ms	1.83 J	83.77%	2.15 GB	$OPQ=256$, $IVF=32768$, $nprobe=32768$
Single Index Default	IVFPQ	13 - 17 ms	0.60 - 0.78 J	28.41%	428 MB	$m=48$, $nprobe=8$, $nlist=4096$
	HNSW	24 - 28 ms	0.96 - 1.2 J	66.27%	22.05 GB	$M=32$, $efC=200$, $efSearch=128$
	KNN (GPU)	51 - 55 ms	6.4 - 7.6 J	100%	21.25 GB	Full Precision
Ensemble	IVFPQ	42–46 ms	1.4–1.6 J	–	–	$m=48$, $nlist=4096$, $nprobe=8$
	HNSW	65–70 ms	2.2–2.4 J	–	–	$M=32$, $efC=200$, $efSearch=128$
Cross-encoder reranker	IVFPQ + BLIP2	4.32 – 4.66 s	353.9 – 390.7 J	–	–	$m=48$, $nprobe=8$, $nlist=4096$
	IVFPQ + Qwen	1.09 – 1.12 s	178.7 – 186.6 J	–	–	$m=48$, $nprobe=8$, $nlist=4096$
	HNSW + BLIP2	4.59 – 4.63 s	367.1 – 386.5 J	–	–	$M=32$, $efC=200$, $efSearch=128$
	HNSW + Qwen	1.10 – 1.12 s	180.9 – 187.7 J	–	–	$M=32$, $efC=200$, $efSearch=128$

Table 8: Unified comparison of retrieval pipelines across latency, per-query energy, and accuracy. Latency and energy values are taken from measured runs; accuracy corresponds to candidate fidelity on LAION-7M.

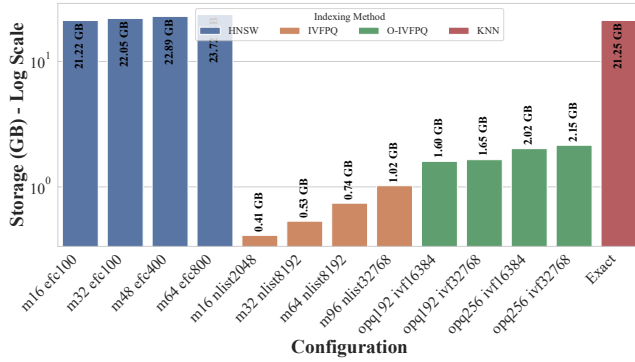


Figure 5: On-disk storage (GB, log scale) for twelve index configurations on LAION-7M across HNSW, IVFPQ, O-IVFPQ, and FLAT-Index (Exact) search.

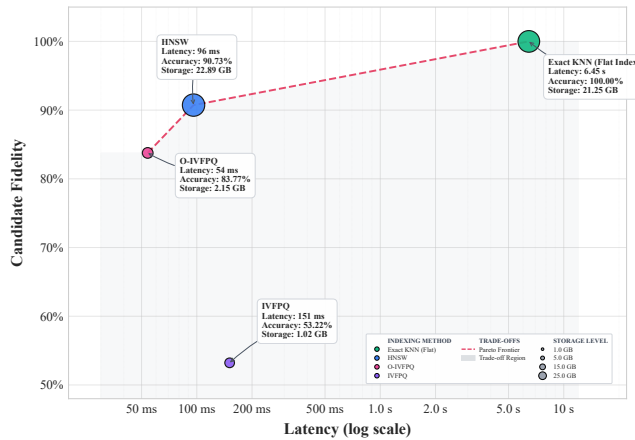


Figure 6: Latency-Fidelity tradeoff for single-index operating points on LAION-7M. The marker size indicates size of index. The curve highlights the index operating points among the evaluated configurations.

fidelity-optimised configurations of Table 7, which use substantially more aggressive search parameters than the defaults ($nprobe=4096$ vs. 8 for IVFPQ; $efSearch=2048$ vs. 128 for HNSW). The resulting latency increase relative to Section 5.2.1 – roughly $4\times$ for HNSW (96 ms vs. 24–28 ms) and $9\text{--}12\times$ for IVFPQ (151 ms vs. 13–17 ms) – reflects the cost of buying the accuracy reported in Table 7, and is the central trade-off this section quantifies. Ensemble and reranker rows continue to use the default configurations and remain directly comparable to Section 5.2.

The following pipeline-level trends emerge directly from Table 8.

O-IVFPQ is the practical sweet spot for single-index retrieval. At 83.77% fidelity, O-IVFPQ trails HNSW by ~ 7 points but runs at 54ms (vs. 96ms), 1.83J (vs. 3.15J), and uses 2.15GB of storage versus HNSW’s 22.89GB.

Standard IVFPQ sacrifices too much candidate fidelity. Standard IVFPQ achieves the smallest footprint but loses substantial candidate fidelity, limiting downstream recovery.

Flat Index is impractical at 7 M scale. Exact Flat search serves primarily as an accuracy reference because of its high latency and storage cost at 7M scale.

For ensembles, IVFPQ is $\sim 1.5\times$ cheaper than HNSW. IVFPQ ensembles run in 42–46 ms at 1.4–1.6J; HNSW ensembles take 65–70 ms at 2.2–2.4J. We do not measure LAION-scale R@1 for ensembles, so the candidate fidelity implications of using IVFPQ carry over from the single-index numbers.

Index choice is irrelevant once a cross-encoder is added. Once reranking is introduced, latency and energy are dominated by the reranker rather than the ANN index.

Qwen3-VL offers most of BLIP-2’s accuracy at $\sim 4\times$ less latency and $\sim 2\times$ less energy. Qwen3-VL reduces latency and energy substantially relative to BLIP-2 while retaining competitive retrieval effectiveness.

Figure 6 shows the single-index operating points from Table 8. Exact flat search has perfect candidate fidelity, but has high latency and high storage cost. As discussed earlier, it is infeasible for large-scale retrieval. IVFPQ is compact, but its low candidate fidelity hurts downstream reranking or fusion. HNSW provides the best fidelity, but at high storage cost. O-IVFPQ occupies the most attractive

intermediate point. It has relatively high fidelity while also having lower latency, energy and storage compared to the other options.

5.3.3 Deployment Recommendations. Translating the findings above into concrete guidance, we highlight the following as a starting point. Any deployment may weigh latency, energy, storage and fidelity differently, based on the resource constraints.

- (1) **Use O-IVFPQ for single-index retrieval.** At our fidelity-optimised operating point it delivers lower latency (54 ms vs. 96 ms), energy (1.83 J vs. 3.15 J), and storage (2.15 GB vs. 22.89 GB) than HNSW, conceding only ~ 7 points of fidelity. Use HNSW for cases where high candidate fidelity is required and where storage is less constrained; reserve standard IVFPQ for cases where aggressive compression matters more than candidate fidelity.
- (2) **Use IVFPQ rather than HNSW for ensembles when under tight latency and energy budgets.** IVFPQ ensembles are $\sim 1.5\times$ faster and consume $\sim 1.5\times$ less energy than HNSW ensembles at default settings, with no additional implementation complexity.
- (3) **Pick the reranker by latency budget, not the index.** With either reranker, IVFPQ and HNSW yield indistinguishable end-to-end latency and energy. Use Qwen3-VL when low latency and energy is required.
- (4) **Invest in the reranker, not the retriever once sufficient candidate fidelity is reached.** Section 5.1 shows that with a strong cross-encoder, diverse first-stage retrievers (image-based, text-based, instruction-tuned) cluster into a narrow R@1 band on MS-COCO. Choose the first stage on efficiency grounds and spend the compute budget on the reranker. Further, Section 5.1 also shows that rerankers reduce sensitivity to the choice of first-stage retriever, provided the relevant item is present in the candidate set.
- (5) **Candidate fidelity is a key deployment metric for ANN-based retrieve-and-rerank or ensemble systems.** Downstream rerankers can reorder retrieved candidates, but cannot recover relevant items if they are omitted during ANN retrieval.

6 Related Work

Our work sits at the intersection of multimodal retrieval modeling, approximate nearest-neighbour search, and system-level evaluation of retrieval pipelines. We review each thread in turn.

Vision-language models for retrieval. The set of Transformer-based multimodal retrieval models parallels the taxonomy well-known from BERT-based text IR: bi-encoders (CLIP [24], FLAVA [28]) emphasize scalability, cross-encoders (BLIP-2 [16]) emphasize accuracy, late-interaction models (LightningDOT [30], llama-nemoretriever-colembed [32], preFLMR [21], ColPali [7]) balance the two, and instruction-tuned retrievers (UniIR [31], InstructBLIP [4]) emphasize flexibility and generalization. Many other models such as [6, 8, 11, 17], fall within these paradigms. For our study, we have selected representative models from each category.

Pipeline-centric evaluation in text IR. The systematic evaluation of multi-stage retrieval pipelines is well established in text IR. Lin et al. [19] survey the use of pretrained transformers in

text ranking and distinguish between first-stage retrieval, reranking, and multi-stage pipelines. Nogueira and Cho [23] demonstrate that BERT-based cross-encoders dramatically improve passage reranking, establishing the retrieve-and-rerank paradigm. The MS MARCO leaderboard and TREC Deep Learning tracks have made pipeline-level comparison standard practice in text IR. However, comparable pipeline-centric studies in the multimodal setting remain scarce. Notable exceptions include ensemble and fusion approaches such as MMMORRF [25], which explores multimodal multilingual fusion strategies using RRF. To the best of our knowledge, no existing work provides a systematic comparison of retrieval pipeline types *jointly with* large-scale efficiency measurement for text-to-image retrieval – a gap we address.

Approximate nearest-neighbour search and indexing. Large-scale dense retrieval requires efficient nearest-neighbour search. HNSW [22] constructs a hierarchical navigable small-world graph that supports sub-linear query-time lookups, achieving excellent latency at the cost of large memory consumption. Product quantisation [10] compresses high-dimensional vectors into compact codes, and when combined with inverted-file indexing (IVFPQ), achieves aggressive storage and energy savings at the cost of approximate results. GPU-accelerated brute-force (Flat) search offers exact results with competitive latency via parallelism but poor storage efficiency. These indexing strategies have been extensively evaluated for text retrieval, but their interaction with multimodal encoders—particularly in multi-stage pipelines—remains under-studied.

Energy and sustainability in retrieval. Studies on the carbon cost of NLP models [29] have raised awareness of computational sustainability, and Scells et al. [26] proposed a framework for minimizing energy usage in IR research. However, our work is among the first to provide fine-grained energy breakdowns across pipeline stages and indexing strategies for cross-modal retrieval.

7 Conclusions

This paper presented a pipeline-centric evaluation of cross-modal text-to-image retrieval, jointly analysing accuracy and efficiency. First, pipeline composition has a substantial impact on retrieval effectiveness. Retrieve-and-rerank pipelines consistently outperform standalone retrieval systems on R@1, while ensemble pipelines remain competitive at higher recall levels. Our results further show that once the relevant item is reliably present in the candidate set, downstream rerankers become less sensitive to the specific choice of first-stage retriever. Second, at 7M scale, O-IVFPQ provides the best single-index operating point among the evaluated approximate indexes. Third, reranker choice is as important as index choice for system design. Qwen3-VL achieves $\sim 4\times$ lower latency and $\sim 2\times$ lower energy than BLIP-2, with only a modest reduction in R@1 for strong first-stage retrievers. This makes retrieve-and-rerank pipelines feasible for sub-1.5 s applications that would be impractical with BLIP-2. Together with low-latency IVFPQ ensembles and highly efficient single-index retrieval, these results reveal operating points spanning different accuracy–efficiency trade-offs.

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GenAI Usage Disclosure

Generative AI tools were used to assist with writing and debugging experiment code, and editing the text of this paper.

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